

ExampleJupyterNotebookReport

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1 Example Solution to Homework 1

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You might want to do your entire analysis and report using a jupyter notebook.

This example notebook reproduces the content of the latex example and shows how to make a simple graph.

2 Problem 1: Features of a Jupyter Notebook (40 points)

Here, you will see how to write mathematical formulas, include code, and show data sets in a Jupyter notebook. This cell (and those below) are Markdown cells. You can double-click any markdown cell to how it was written.

2.0.1 Including math

Write your solution here. If there were math to write, you can use latex commands. Jupyter will translate those into nice equations.

$$\begin{aligned}\sin A \cos B &= \frac{1}{2} [\sin(A - B) + \sin(A + B)] \\ \sin A \sin B &= \frac{1}{2} [\cos(A - B) - \cos(A + B)] \\ \cos A \cos B &= \frac{1}{2} [\cos(A - B) + \cos(A + B)]\end{aligned}$$

2.0.2 Including code

If you need to include a block of code written in C/C++:

```
// Some interesting code here
int a = 0;
int b = 1;

for(int i=0; i< 10; i++) {
    printf("hello\n")
}
```

or show some command run in the shell

```
>>> ls -a
>>> pwd
```

or show python code

```
a = 0
b = 1
for i in range(10)
    print("hello")
```

The above examples use syntax highlighting. Notice that by specifying the language (C/C++, Python, or Bash), special keywords (e.g., “for” in the for-loop) show up with highlighted colors.

If the code had output, we could include it like so:

```
[ 0.0156615   0.00483956  0.00490249  0.00701772  0.0049713   0.00542965
 0.00486095  0.00420872  0.02011727]
```

2.0.3 Including a figure

The supercomputer Blue Waters has hundreds of thousands of CPUs. A picture of Blue Waters can be included as follows



notice that the image file “BlueWaters.jpg” needs to be in the same folder as this notebook.

```
[8]: # Or you can include a figure by executing this python code
from IPython import display

display.Image("BlueWaters.jpg") # This can display many different kinds of
↪ images
```

[8]:



3 Problem 2: analysis of a data file (50 points)

Scott's CPU monitoring script (task 1 of lab 2) produced a data file called CPUData.dat

In this section, we will see how Python code can be easily included in the notebook for quick data analysis. Unlike the above example, we will not execute the python code. This works because the following cells are “code cells”, whereas the above ones were markdown.

```
[10]: import numpy as np
from matplotlib import pyplot as plt

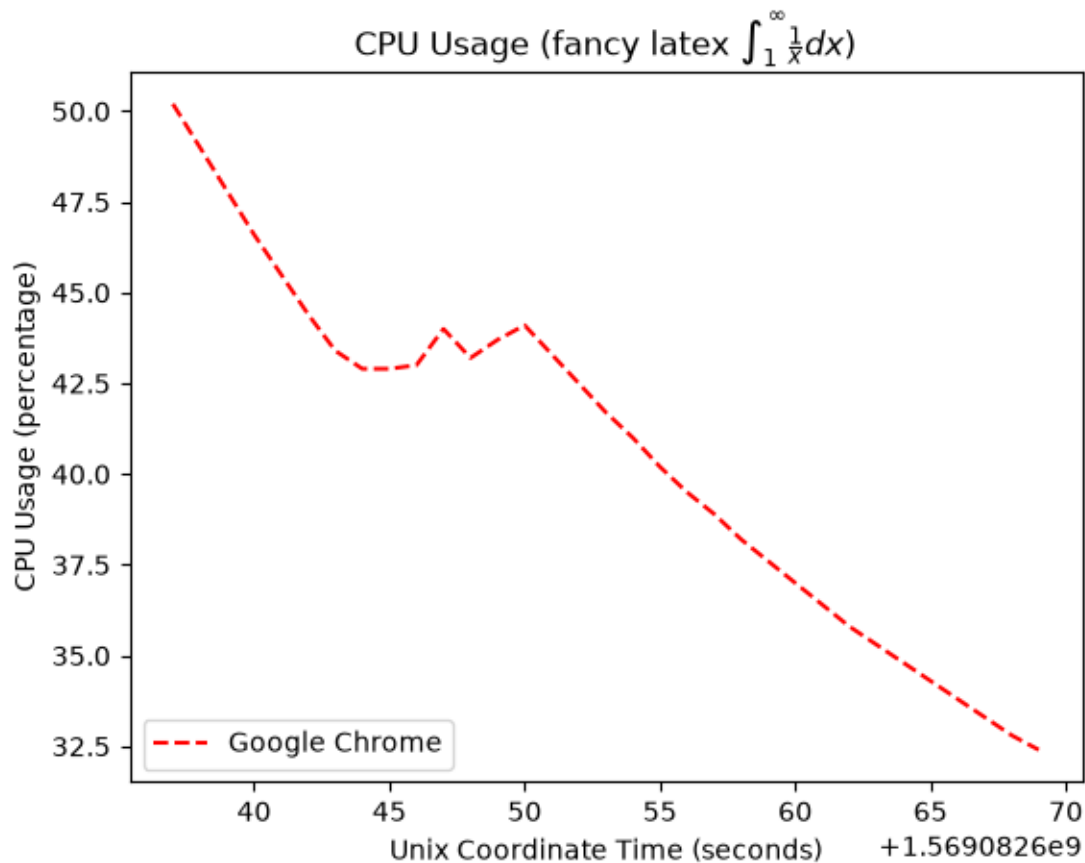
data = np.loadtxt('CPUData.dat') # <- Will automatically ignore lines starting
    ↪with #

x = data[:,0]
y = data[:,1]

plt.figure()
plt.plot(x,y,'r--',label='Google Chrome')
plt.xlabel('Unix Coordinate Time (seconds)')
plt.ylabel('CPU Usage (percentage)')

# NOTE: Notice we are putting a latex equation into the title
# by using "raw strings" (notice the small "r" in front of the string)
plt.title(r'CPU Usage (fancy latex  $\int_1^{\infty} \frac{1}{x} dx$ )')

plt.legend(loc=3)
plt.show()
# The next line is commented out. Uncomment it if you'd like to save the image
    ↪as a png
#plt.savefig("cpu_vs_time.png")
```



If this were a real homework report, you would then provide some descriptive text here describing the figure and its importance in some detail.

4 Problem 3: using jupyter notebook as a shell terminal (50 points)

It turns out Jupyter Notebooks also support shell-like commands. Lets try some out

- cat code.c for including code into jupyter notebooks

pwd

[11]: ls

BlueWaters.jpg* CPUDData.dat*
ExampleJupyterNotebookReport.ipynb

In this next example, we need to use Jupyter notebook “magic” command “%%bash”. This will run the block of commands after it inside of your shell.

This is useful for more complicated tasks.

Here we create a simple C program using echo. you should have a new file called “hello.c” create in this directory.

```
[12]: %%bash
echo '#include <stdio.h>
int main() {
    printf("Hello, World!\\n");
    return 0;
}' > hello.c
```

Now we can use the Unix command Cat to show the contents of the new hello.c file directly inside of the notebook.

This is another convenient way to include code into your Jupyter notebook.

```
[13]: cat hello.c

#include <stdio.h>
int main() {
    printf("Hello, World!\\n");
    return 0;
}
```

4.1 Final step... export as a PDF

You can turn your notebook into a PDF by doing

File -> Save and Export Notebook As...

from the menu. If this were a real homework, you would turn in the PDF in class while also uploading the Jupyter notebook to bitbucket.

```
[ ]:
```